HALEY SAPP

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EDUCATION

University of North Carolina Wilmington | B.S. Computer Science, Artificial Intelligence Specialization | Expected Dec 2026
Forsyth Technical Community College | Associate of Science, May 2024 | Phi Theta Kappa

TECHNICAL SKILLS

Programming: Python, Java, C++, Swift, SQL, HTML, SML, LC-3 Assembly
AI / Computer Vision: YOLO, OpenCV, optical flow, K-means, KNN, image processing, LLMs, RAG, model evaluation
Development Tools: PySide6/Qt, SwiftUI, StoreKit 2, NumPy, Pandas, Matplotlib, Git, GitHub, Xcode, VS Code, Jupyter

SELECTED PROJECTS

Hurricane Eye Detection & Tracking | Honors Research | Python, YOLO, OpenCV, PySide6

- Developed a Python computer vision pipeline combining tiled YOLO detection, temporal tracking, optical flow, and rotation analysis on GOES-16 infrared satellite imagery.
- Curated 397 satellite images from 10 hurricanes; achieved 0.675 mAP@50 on held-out Milton test tiles and tracked the eye in 34 of 39 Milton frames (87.2%).
- Co-developed a PySide6 desktop interface presenting eye coordinates, velocity, rotation, confidence, image overlays, and CSV export. Demo: youtu.be/7Wk1g508894

MirrorStat TCG | iOS / iPadOS App | Swift, SwiftUI, StoreKit 2

- Built an iPhone/iPad tabletop card-game companion with configurable stat tracking, turn-based and persistent effects, saved layouts, and iPad two-player mode.
- Implemented StoreKit 2 premium feature gating and published the application on the Apple App Store. App Store: apps.apple.com/us/app/mirrorstat-tcg/id6812886410

Hurricane Safety AI Chatbot | LLMs, Retrieval-Augmented Generation

- Built a hurricane-preparedness chatbot exploring local language models, retrieval-augmented generation, knowledge-base design, prompt engineering, and response evaluation. Demo: youtu.be/MwbDR18hw-4

LEADERSHIP & AEROSPACE EXPERIENCE

NASA Community College Aerospace Scholars (NCAS) | 2023–2024

- Completed Missions 1–2 and was selected among 100 students nationwide for Mission 3 at NASA Armstrong Flight Research Center.
- Collaborated on aerospace engineering concepts involving eVTOL aircraft, battery management, aerodynamics, and flight-control systems.